

Spherical Harmonics

Scalar, vector, and tensor spherical harmonics.

1 Recap: Scalar Spherical Harmonics (Y_{lm})

- **Definition:** Functions $Y_{lm}(\theta, \phi)$ defined on the surface of a unit sphere (S^2).
- **Properties:**
 - They are eigenfunctions of the angular part of the Laplacian (Laplace-Beltrami operator ∇_S^2) and the z-component of the angular momentum operator (L_z):

$$\nabla_S^2 Y_{lm} = -l(l+1)Y_{lm}$$

$$L_z Y_{lm} = mY_{lm}$$

- l is the total angular momentum quantum number ($l = 0, 1, 2, \dots$), m is the azimuthal quantum number ($-l \leq m \leq l$).
- They form a **complete orthonormal basis** for square-integrable scalar functions $f(\theta, \phi)$ on the sphere:

$$f(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_{lm} Y_{lm}(\theta, \phi)$$

where $f_{lm} = \int_{S^2} f(\theta, \phi) Y_{lm}^*(\theta, \phi) d\Omega$ ($d\Omega = \sin \theta d\theta d\phi$ is the solid angle element).

- **Parity:** Under the parity operation ($\mathbf{r} \rightarrow -\mathbf{r}$, which means $\theta \rightarrow \pi - \theta$, $\phi \rightarrow \phi + \pi$), Y_{lm} transforms as $Y_{lm}(\pi - \theta, \phi + \pi) = (-1)^l Y_{lm}(\theta, \phi)$. They have definite parity $(-1)^l$.

2 Vector Spherical Harmonics (VSHs)

Now, consider a *vector field* $\mathbf{V}(\theta, \phi)$ defined on the sphere. How can we expand it? Simply multiplying Y_{lm} by constant basis vectors ($\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$) isn't natural on the sphere. Multiplying by spherical basis vectors ($\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi$) is problematic because $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$ change direction with position.

We need *vector basis functions* defined on the sphere that transform irreducibly under rotations (i.e., have definite total angular momentum j and projection m) and definite parity.

Construction Strategy

Combine the known scalar harmonics Y_{jm} with vector operators that act on the sphere in a way that preserves rotational symmetry properties. The key operators are:

- Radial unit vector: $\mathbf{n} = \mathbf{e}_r$
- Surface gradient: $\nabla_S = \mathbf{e}_\theta \frac{\partial}{\partial \theta} + \mathbf{e}_\phi \frac{1}{\sin \theta} \frac{\partial}{\partial \phi}$
- Angular momentum operator (vector): $\mathbf{L} = -i(\mathbf{r} \times \nabla) = -i \left(\mathbf{e}_\phi \frac{\partial}{\partial \theta} - \mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right)$

From these, we can construct three fundamental types of VSHs for a given (j, m) (note: often l is used instead of j , I'll use j here for clarity when distinguishing from scalar l):

- **Type 1: Radial (or Longitudinal)**
 - $\mathbf{V}_{jm}^{(r)} = \mathbf{e}_r Y_{jm}(\theta, \phi)$
 - This represents a purely radial vector field whose angular distribution is given by Y_{jm} .

- Exists for $j \geq 0$.

- **Type 2: Electric (or Polar, Gradient type)**

- $\mathbf{Y}_{jm} = \frac{1}{\sqrt{j(j+1)}} \nabla_S Y_{jm}(\theta, \phi)$
- This vector field is purely *tangential* to the sphere (no $\mathbf{e}_r$ component). It's constructed from the gradient of a scalar potential Y_{jm} . The normalization factor $[j(j+1)]^{-1/2}$ is conventional.
- Since $\nabla_S Y_{00} = 0$, this type exists only for $j \geq 1$.
- It represents fields like the tangential part of an electric field derived from a scalar potential with angular dependence Y_{jm} .

- **Type 3: Magnetic (or Axial, Curl type)**

- $\mathbf{\Psi}_{jm} = \frac{1}{\sqrt{j(j+1)}} \mathbf{L} Y_{jm}(\theta, \phi) = \mathbf{e}_r \times \mathbf{Y}_{jm}$
- This vector field is also purely *tangential* to the sphere. It's related to the curl of a radial vector potential (or gradient of a pseudo-scalar). The second equality ($\mathbf{e}_r \times \mathbf{Y}_{jm}$) shows it's perpendicular to both the radial direction and the 'Electric/Polar' VSH.
- Since $\mathbf{L} Y_{00} = 0$, this type also exists only for $j \geq 1$.
- It represents fields like a magnetic field generated by radial currents, or the tangential part of a field derived from a vector potential $\mathbf{A} = \mathbf{e}_r a(r) Y_{jm}$.

Properties of VSHs

- **Completeness:** Any reasonably smooth vector field $\mathbf{V}(\theta, \phi)$ on the sphere can be uniquely expanded as:

$$\mathbf{V}(\theta, \phi) = \sum_{j=0}^{\infty} \sum_{m=-j}^j \left[f_{jm}(r) \mathbf{V}_{jm}^{(r)} + a_{jm}(r) \mathbf{Y}_{jm} + b_{jm}(r) \mathbf{\Psi}_{jm} \right]$$

(If the field is defined only *on* the sphere, the coefficients f, a, b are constants. If it's a 3D field evaluated on spherical shells, they depend on radius r). For purely tangential fields, $f_{jm} = 0$.

- **Orthogonality:** The VSHs form an orthogonal set over the sphere:

$$\begin{aligned} \int_{S^2} \mathbf{V}_{j'm'}^{(r)*} \cdot \mathbf{V}_{jm}^{(r)} d\Omega &= \delta_{jj'} \delta_{mm'} \\ \int_{S^2} \mathbf{Y}_{j'm'}^* \cdot \mathbf{Y}_{jm} d\Omega &= \delta_{jj'} \delta_{mm'} \quad (\text{for } j, j' \geq 1) \\ \int_{S^2} \mathbf{\Psi}_{j'm'}^* \cdot \mathbf{\Psi}_{jm} d\Omega &= \delta_{jj'} \delta_{mm'} \quad (\text{for } j, j' \geq 1) \end{aligned}$$

Integrals involving products of *different types* of VSHs are zero.

- **Parity:** This is crucial for Regge-Wheeler.

- $\mathbf{e}_r$ changes sign under parity ($\mathbf{e}_r \rightarrow -\mathbf{e}_r$). Y_{jm} gives $(-1)^j$. $\text{Parity}(\mathbf{V}_{jm}^{(r)}) = (-1) \times (-1)^j = (-1)^{j+1}$
- ∇_S is a *polar* vector operator (doesn't introduce a sign change under parity itself). $\text{Parity}(\mathbf{Y}_{jm}) = (+1) \times (-1)^j = (-1)^j$ (**Even/Polar Parity**)
- $\mathbf{L} = -i(\mathbf{r} \times \nabla)$ is an *axial* vector operator (pseudo-vector, introduces a sign change under parity). $\text{Parity}(\mathbf{\Psi}_{jm}) = (-1) \times (-1)^j = (-1)^{j+1}$ (**Odd/Axial Parity**)

Summary for VSHs

We have decomposed vector fields on the sphere into radial parts and tangential parts. The tangential parts further split into two types based on how they are generated (gradient-like or curl-like) which correspond directly to their parity properties: $\mathbf{Y}_{jm}$ have "natural" parity $(-1)^j$ (Polar/Even), while $\mathbf{\Psi}_{jm}$ have "unnatural" parity $(-1)^{j+1}$ (Axial/Odd).

3 Tensor Spherical Harmonics (TSHs)

Now, we extend this to rank-2 tensor fields $\mathbf{T}(\theta, \phi)$ on the sphere, like the metric perturbation $h_{\mu\nu}$ restricted to the sphere in the Regge-Wheeler paper. We need basis *tensor fields* with definite angular momentum (j, m) and parity.

Construction Strategy

Combine the scalar Y_{jm} , the VSHs ($\mathbf{V}^{(r)}, \mathbf{Y}, \mathbf{\Psi}$), and invariant tensors on the sphere (like the metric tensor $\mathbf{g}_S = \mathbf{e}_\theta \otimes \mathbf{e}_\theta + \mathbf{e}_\phi \otimes \mathbf{e}_\phi$) using operations like the tensor product ($\otimes$) and the surface covariant derivative (∇_S).

Again, the most physically meaningful classification is based on **parity**. We construct TSHs that are either purely Even/Polar or purely Odd/Axial. We are often interested in *symmetric* rank-2 tensors.

Even Parity (Polar) TSHs (Parity $(-1)^j$)

These are constructed using the scalar Y_{jm} and the *polar* VSH $\mathbf{Y}_{jm}$.

- **Type E0 (Scalar-derived, Trace):**

$$\mathbf{T}^{(jm)\text{E0}} = Y_{jm} \mathbf{g}_S$$

This is proportional to the metric tensor on the sphere. It's symmetric. Exists for $j \geq 0$.

- **Type E1 (Vector-derived, Radial-Tangential):** (Often needed for full 3+1 decomposition, but let's focus on tangential parts relevant to Regge-Wheeler's 2D sphere analysis)

$$\sim (\mathbf{V}_{jm}^{(r)} \otimes \mathbf{Y}_{jm} + \mathbf{Y}_{jm} \otimes \mathbf{V}_{jm}^{(r)})$$

Symmetric combination mixing radial and tangential (polar) vectors.

- **Type E2 (Tensor-derived, Tangential Symmetric Trace-Free):**

$$\mathbf{T}^{(jm)\text{E2}} \sim \left(\nabla_S \otimes \nabla_S - \frac{1}{2} \mathbf{g}_S \nabla_S^2 \right) Y_{jm}$$

This is constructed by taking two surface covariant derivatives of Y_{jm} and making it symmetric and trace-free. It is purely tangential. (The $\nabla_S^2 Y_{jm}$ term effectively removes the trace part proportional to $\mathbf{g}_S$). It exists for $j \geq 2$. Parity comes from Y_{jm} , so it's $(-1)^j$.

Odd Parity (Axial) TSHs (Parity $(-1)^{j+1}$)

These are constructed using the *axial* VSH $\mathbf{\Psi}_{jm}$.

- **Type O1 (Vector-derived, Radial-Tangential):**

$$\sim (\mathbf{V}_{jm}^{(r)} \otimes \mathbf{\Psi}_{jm} + \mathbf{\Psi}_{jm} \otimes \mathbf{V}_{jm}^{(r)})$$

Symmetric combination mixing radial and tangential (axial) vectors.

- **Type O2 (Tensor-derived, Tangential Symmetric Trace-Free):**

$$\mathbf{T}^{(jm)O2} \sim (\nabla_S \otimes \Psi_{jm} + \Psi_{jm} \otimes \nabla_S)_{\text{symm, trace-free}}$$

(Symmetrized covariant derivative of the axial VSH, made trace-free). This tensor is purely tangential, symmetric, and can be made trace-free. It exists for $j \geq 2$. Parity comes from Ψ_{jm} , so it's $(-1)^{j+1}$.

(Note: Precise definitions and normalizations for TSHs can vary between sources. The key concept is the construction principle and the resulting parity.)

Properties of TSHs

- **Completeness:** Any sufficiently smooth rank-2 tensor field on the sphere can be expanded using a complete set of TSHs (including radial-radial, radial-tangential, and tangential-tangential types, often further decomposed into symmetric/anti-symmetric and trace/trace-free parts).
- **Orthogonality:** Properly defined TSHs form an orthogonal basis with respect to a suitable inner product for tensors on the sphere.
- **Parity:** This is the most critical property for the application. Tensors built from polar objects ($Y_{jm}, \mathbf{Y}_{jm}, \mathbf{g}_S, \nabla_S$) have even parity $(-1)^j$. Tensors involving an odd number of axial objects (Ψ_{jm}) have odd parity $(-1)^{j+1}$.

4 Relevance to Regge-Wheeler (Stability of Schwarzschild)

1. **The Problem:** They studied linear metric perturbations ($h_{\mu\nu}$) around the static, spherically symmetric Schwarzschild background metric ($g_{\mu\nu}^{(0)}$).
2. **Symmetry:** Because the background is spherically symmetric, it makes sense to decompose the perturbation $h_{\mu\nu}$ using basis functions that respect this symmetry – the TSHs (and VSHs/Scalars for different components).
3. **Decomposition:** They expressed the components of $h_{\mu\nu}$ in a specific gauge (Regge-Wheeler gauge) as sums over modes (l, m) involving scalar, vector, and tensor spherical harmonics. For instance, the tangential part of the spatial metric perturbation h_{ab} (where $a, b = \theta, \phi$) can be written as:

$$h_{ab}(t, r, \theta, \phi) = \sum_{l, m} \left[H_{lm}^{(0)}(t, r) \mathbf{T}_{ab}^{(lm)E0} + H_{lm}^{(2)}(t, r) \mathbf{T}_{ab}^{(lm)E2} + J_{lm}(t, r) \mathbf{T}_{ab}^{(lm)O2} \right]$$

(Similar expansions hold for other components like h_{tr}, h_{rr}, h_{ta} involving VSHs and scalar Y_{lm}).

4. **The Crucial Decoupling:** The power of this method, combined with a clever gauge choice, is that the linearized Einstein equations *decouple* based on parity. All the perturbation components associated with **Odd/Axial Parity** TSHs (like $\mathbf{T}^{(lm)O2}$) evolve independently from the components associated with **Even/Polar Parity** TSHs (like $\mathbf{T}^{(lm)E0}, \mathbf{T}^{(lm)E2}$). Furthermore, modes with different (l, m) also decouple due to linearity and orthogonality.
5. **Master Equations:** This decoupling allowed Regge and Wheeler to derive a single *scalar* wave equation for a master function $\Phi_{lm}^{(\text{odd})}(t, r)$ that encapsulates the entire dynamics of the odd/axial perturbations for a given (l, m) . This is the famous **Regge-Wheeler equation**. Later, Zerilli found the corresponding master equation for the even/polar perturbations.
6. **Stability Analysis:** By analyzing the potential term in the Regge-Wheeler (and later Zerilli) equation, they could show that there were no exponentially growing modes (instabilities). All perturbations would either disperse as gravitational waves or fall into the black hole.

In Summary

- Scalar Harmonics (Y_{lm}): Basis for scalar functions on S^2 , parity $(-1)^l$.
- Vector Harmonics ($\mathbf{V}^{(r)}, \mathbf{Y}, \mathbf{\Psi}$): Basis for vector fields on S^2 . $\mathbf{Y}$ (polar/even) has parity $(-1)^j$, $\mathbf{\Psi}$ (axial/odd) has parity $(-1)^{j+1}$.
- Tensor Harmonics ($\mathbf{T}^E, \mathbf{T}^O$): Basis for tensor fields on S^2 . Even/Polar ($\mathbf{T}^E$) built from $Y_{jm}, \mathbf{Y}_{jm}$ have parity $(-1)^j$. Odd/Axial ($\mathbf{T}^O$) built using $\mathbf{\Psi}_{jm}$ have parity $(-1)^{j+1}$.
- Regge-Wheeler used this framework to decompose metric perturbations. The decoupling of odd and even parity modes under linearized gravity around Schwarzschild was the key simplification that allowed them to prove stability by analyzing separate master scalar wave equations.

This framework is fundamental not just for black hole stability but also for calculating gravitational waveforms from sources involving black holes (like mergers and ringdown) and other problems in physics involving fields on a sphere. For deeper study, consult advanced GR textbooks (Wald, Poisson & Will) or specialized reviews on black hole perturbation theory or gravitational waves (e.g., by Thorne, Kokkotas & Schmidt). Jackson's E&M textbook also has a good section on VSHs in the context of multipole radiation.